

Lab: Entity Extraction and Memory Storage using AI Builder and Power Automate

Objective

In this lab, you will:

- Extract entities from unstructured text using AI Builder.
- Identify entities such as Person Name, Organization, City, and Email.
- Store extracted entities in an Excel table.
- Build a simple long-term memory repository that can be used for Retrieval-Augmented Generation (RAG) scenarios.

Prerequisites

- Microsoft Power Automate
- AI Builder Credits
- OneDrive for Business
- Excel Online

Step 1: Create Excel Memory Store

Create a new Excel file:

Entity Capture June.xlsx

Add the following columns:

EntityType	EntityValue	Confidence	CreatedDate
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Select the data range and convert it into an Excel Table.

Rename the table:

EntityMemory

Save the workbook in OneDrive.

Step 2: Create a New Flow

Create an Instant Cloud Flow.

Select:

Manually trigger a flow

Add a Text Input:

text

This input will receive the text that needs to be analyzed.

Step 3: Add AI Builder Entity Extraction

Add action:

Extract standard entities

Input:

Text = Text input from Manual Trigger

The AI Builder model will automatically identify:

- Person Names
- Organizations
- Cities
- Emails
- Phone Numbers
- Addresses
- Dates

depending on the input text.

Step 4: Create Apply to Each Loop

Add:

Apply to each

Select output:

Expression:

```
outputs('Extract_standard_entities')?['body']?['responsev2']?  
['predictionOutput']?['result']?['entities']
```

This loop will iterate through every entity identified by AI Builder.

Step 5: Store Entities in Excel

Inside the Apply to Each loop add:

Add a row into a table

Configure:

Location:

OneDrive for Business

Document Library:

OneDrive

File:

Entity Capture June.xlsx

Table:

EntityMemory

Step 6: Map Excel Columns

EntityType

Expression:

```
items('Apply_to_each')['type']
```

EntityValue

Expression:

```
items('Apply_to_each')['value']
```

Confidence

Expression:

```
items('Apply_to_each')['score']
```

CreatedDate

Expression:

```
utcNow()
```

Final Flow

```
Manually Trigger a Flow
↓
Extract Standard Entities
↓
Apply to Each Entity
↓
Add Row into Excel Table
```

Test Input

```
My name is Abdullah Khan.
I work at Microsoft.
I live in Hyderabad.
My email is abduallahkhan@microsoft.com.
```

Expected AI Output

Entity Type	Entity Value
PersonName	Abdullah Khan
Organization	Microsoft
City	Hyderabad
Email	abduallahkhan@microsoft.com

Expected Excel Output

EntityType	EntityValue	Confidence	CreatedDate
PersonName	Abdullah Khan	0.99	Current Date
Organization	Microsoft	0.90	Current Date
City	Hyderabad	0.90	Current Date

EntityType	EntityValue	Confidence	CreatedDate
Email	abdullahkhan@microsoft.com	0.80	Current Date

Outcome

At the end of this lab, participants will be able to:

- Extract entities from free-form text.
- Process AI Builder outputs.
- Loop through extracted entities.
- Persist entity data into Excel.
- Build a simple memory layer that can later be used for RAG and AI agent scenarios.